Int J Syst Assur Eng Manag
https://doi.org/10.1007/s13198-025-02775-8

ORIGINAL ARTICLE

A TOTP-based secure data storage system in the cloud environment using the JWT token approach

Shahnawaz Ahmad[1] · **Mohd Arif**[2] · **Javed Ahmad**[3] · **Shabana Mehfuz**[4]

Received: 14 December 2023 / Revised: 16 November 2024 / Accepted: 22 March 2025
© The Author(s) under exclusive licence to The Society for Reliability Engineering, Quality and Operations Management (SREQOM), India and The Division of Operation and Maintenance, Lulea University of Technology, Sweden 2025

Abstract Cloud computing (CC) facilitates online computing resources on a pay-per-use basis. Recently, due to advancements in internet technologies, the popularity of CC has surged. Various users and organizations are adopting it for application, infrastructure, and platform as a service. Cloud storage is one of the major services that is widely being used by both organizations and individual users. Organizations like healthcare, governance, finance, defense etc., are also using the cloud infrastructure to store their sensitive data and thus making it more vulnerable to security attacks. This work proposes a secure cloud-based storage system called TrustStore that implements a dual security mechanism: one for the data in storage and the other for user authentication to access the data. TrustStore uses a time-based one-time password as the authentication mechanism, which is a time-constrained, unique temporary code generated for the user to gain system access. Additionally, to protect the data in TrustStore, cost-effective RSA and AES algorithms have been implemented to perform the encryption and decryption of data. The work has been implemented over various file types of different sizes and their uploading and downloading time has been calculated. This work also presents a case study in which financial data for a banking system is considered for protection against malicious activities by intruders.

Keywords Cloud computing · One-time password · JSON web token · Security · Key management system

1 Introduction

CC is an intelligent framework facilitating computing resources to users over the Internet. It provides three basic services like application as service, platform as service and infrastructure as services (Laghari et al. 2022). These days CC has gained popularity in various fields like transportation, entertainments, education, agriculture, real state and many more. Recently it also has established its significance in crucial sectors like, healthcare, governance, finance, defense etc., where sensitive data are also being shared from one device to other using the platform. Considering its applicability and popularity in such a diverse field, maintaining the integrity and confidentiality of data is of prime concern for both users and organizations. However, cloud paradigms are not free from the violations affecting data integrity, access control, identity management and confidentiality. The platform has two vulnerable states where data can be compromised: one during transit and the other during storage (Chen 2023; Fatima et al. 2022). Many applications and algorithms have been developed in literature to address both the sites of vulnerability. This study is particularly concerned with the security of data in cloud storage, where users or organizations save their confidential data like medical records,

✉ Shahnawaz Ahmad
shahnawaz98976@gmail.com

[1] School of Computer Science Engineering and Technology, Bennett University, Greater Noida, Uttar Pradesh 201310, India

[2] School of Computer Science & Engineering, Galgotias University, Greater Noida, Uttar Pradesh 203201, India

[3] Department of Computer Science and Engineering, Sharda School of Computing Science and Engineering, Sharda University, Greater Noida 201310, India

[4] Department of Electrical Engineering, Jamia Millia Islamia (A Central University), New Delhi 110025, India

Published online: 15 April 2025

financial information, and military intelligence considering it a safe vault (Laghari et al. 2022; Jumani and Laghari 2021; Laghari et al 2023). The literature has numerous strategies to protect integrity and confidentiality of these data in storage. Multi-cloud appears to be one of the viable solutions for this, where confidentiality and availability of data are improved by dividing the user's data block into data elements and distributing them among the available resources facilitated by Cloud Service Providers (CSPs). This allows a predetermined number of CSPs to help recover the complete data block when it is required by the legitimate users. However, due to the increased sophistication of cyber threats and the growing reliance on cloud services for sensitive information, enhancing security of data in a multi-cloud context has recently become more critical than ever.

Other efficient strategy presents in literature includes encryption of data before it gets stored into the cloud storage (Failed 2014; Cai et al. June 2023, Namasudra et al. 2018). Here, the end device encrypts the data after its arrival from another source and saves it to the cloud. Similarly, the data is decrypted to its original form when legitimate users want to access it. Thus, it requires two levels of protection mechanism: one for the safety of data in storage, and the other for the authentication of legitimate users during data access. This works introduces a hybrid mechanism to cater for the level of protection. To ensure the safety of data in storage, we use RSA and AES algorithms to perform the encryption and decryption of data at the storage level. The minimal cost small AES key has been used for the task.

Coming to user authentication, most online services nowadays use multi-factor authentication, which operates on multi-step verification. Access control to a device is granted to users who provide additional information besides the password and thus making it more difficult for an intruder to pretend to be another user. One of the ways to get additional information from the user may be the One Time Password (OTP) received on the registered mobile number or email address of the user (Islam et al. 2017; Mohamad et al. 2021). Complex mathematical algorithms are used to implement OTP and thus making it a robust cryptographic technique (Rasori et al. 2022). For every access request, the cloud server sends an OTP to the legitimate user. In response, the user must provide the OTP; If the OTP matches accurately, access is granted, otherwise, access is denied. Though OTP is a technological safeguard for illegal access to the account, making it time constraints adds an extra layer of security to it. This implies that the OTP is active only when the specific-time action is in effect. This OTP is called TOTP and considered as less vulnerable than the simple OTP based security mechanism (Idayathulla 2019; Uymatiao and Yu 2014a). Thus, this works implements TOTP as the extra information required by the server for user authentication process.

The significant contributions of the paper can be summarized as given below:

- The work proposes a cloud-based storage architecture called Trust-Store.
- Trust-Store provides two levels of protection mechanism, one for user authentication and the other for storage.
- The user authentication process is implemented using TOTP, and security of data in the storage is ensured by RSA and AES algorithms where minimal cost small AES key have been used.

This paper addresses the challenge of securing data in cloud environments. It starts by reviewing the latest research on this topic in Sect. 2. Section 3 then clarifies the system's authentication process. The heart of the paper lies in Sect. 4, which meticulously details the proposed architecture designed to safeguard cloud data. To understand how well this architecture performs, Sect. 5 presents an evaluation of the system. Section 6 brings the theoretical aspects to life by showcasing a concrete use case for the proposed framework. Finally, Sect. 7 concludes the paper by summarizing the key points and potential implications.

2 Related work

In the past few years, several security solutions have been introduced to bolster the safety of cloud data storage. Many cloud users implement cryptographic techniques to encrypt sensitive information and identify individuals at the other end of the connection to keep information over the Internet as secure as servers can (Xuanzhi and Ahmad 2019; Alam et al. 2020). Community security must pace up with the current threats as nowadays, most personal information is stored virtually (Hassan and Shukur 2019; Aigbe and Akpojaro 2014). Today's vast majority of websites rely more on a traditional verification pattern of providing a username and password than a secure link. Even if the password is needed to confirm the customer's identity, the user's name is utilized to find what online account a client would like to access. Although theoretically, it is secure, several passwords continue to be compromised (Alsayed and Bilgrami 2017; Gualdoni et al. 2017). This is due to weak passwords and gear that can crack passwords more quickly (Uymatiao and Yu 2014b; Hassan et al. 2020). To prevent these assaults, two-factor authentication is a significant approach to secure online transactions, identify the real person, and ensure that only the right person logs into the system or software.

Cryptocurrencies are secured using the time-based authentication of multi-factor tokens. The authors have developed an enhancement approach for MFA Crypto-financial (CR) to provide an additional layer of security. In any 2FA-enabled entity, the user must first submit a login and password before using the token to create a TOTP as a second factor virtually (Taher et al. 2019). A challenge-response protocol for e-commerce based on a concept like TOTP is proposed in. Aina et al. (2018) discussed Scan2-Pass payments for the banking sector in their research study (Aina et al. 2018). In Scan2Pass payments, both the server and the client sides are mandatory for the system to operate. After logging into the server, the user must generate a QR code, along with their username and password. After entering the user login details, the user must run his mobile application on the client side and scan the QR code (Arvind et al. 2020). Several studies have explored using QR codes and Time-based One-Time Passwords (TOTP) to enhance security in financial transactions. Abhishek et al. (2020) proposed a system where a QR code containing a TOTP is scanned by the user's device. The server then verifies if the generated TOTP on the user's device matches the one embedded in the QR code, granting access only if they align. Similarly, Chowdhury et al. (Choudhary et al. 2017) suggested leveraging OTP and QR codes for authentication within online banking systems Chowdhury et al., (Choudhary et al. 2017). These approaches offer an additional layer of security by requiring not just a password but also a temporary code that refreshes frequently.

A study (Bansal and Aggarwal 2024a) introduces a combined Particle Whale Optimization Algorithm (PWOA) for workflow scheduling in cloud–fog computing. By incorporating PSO and WOA it achieves, less execution cost, Makespan and energy consumption than the current approaches for five scientific workflows. The approach also makes it possible to customize the scheduling, this will make the approach more efficient and flexible to the users.

In (Bansal and Aggarwal 2024b), therefore, a new hybrid PSO-WOA algorithm is presented to enable the elimination of PSO's global optima trapping problem and WOA's restricted search space issue. Experiments have been conducted on five scientific workflows using WorkflowSim where proposed method surpasses ACO, PSO, WOA, GA, and BH-FWA methods in terms of Total Execution Cost (TEC) and Total Execution Time (TET) and proved best on Montage and Epigenomics workflows. Overall, these research findings were statistically supported by the tests.

The Particle Whale Optimization Algorithm (PWOA) improves the scheduling of workflows in cloud-fog computing by reducing Total Execution Time (TET) and Total Execution Cost (TEC). When integrated, the PWOA algorithm eliminates the problem of local optima and demonstrates performance superior to both algorithms in experiments encompassing five distinct workflows (Bansal and Aggarwal 2023).

The current commercially used sites are insecure causing an area of concern in terms of sensitive data. Biometric identification and standard password protection for individuals together with RSA encoding can be used with greater effectiveness for various range of network applications on the second level of security (Rani et al. 2019).

Table 1 compares the cloud security options present in the literature based on many factors. More specifically, it compares the four security objectives of confidentiality (C), data access control (DAC), integrity (I), and authentication (A) to a summary of the various algorithms employed by these approaches. Table 1 reveals an essential fact: none of the current methods achieves the four security goals. We also stress the necessity of developing a novel method of storing data securely in a multi-cloud environment multi-cloud setting. Additionally, we contrast our suggested strategy with currently available options in the final column of Table 1. Our qualitative comparison confirms that our solution uses a robust access control mechanism to lower the chance of data leaking. Furthermore, it uses the concept of multi-cloud, TOTP authentication, and symmetric and asymmetric cryptography to provide access control, confidentiality, and integrity for data stored in the cloud (Stallings 2005; May et al.2002; Nechvatal et al. 2001).

3 Research methodology

Data authentication is critical to protect an application's most important components. Passwords, biometrics, token-based authentication, or certificates can all be used to verify users before granting access to resources (Pavlovski et al. 2015). The most convenient options for authentication in web-based applications have long been cookies and server-based authentication. It will, however, be quite challenging to employ server-based authentication when dealing with multi-platform authentication. Because the server must constantly monitor the session, session-based authentication causes server overload and degrades server performance. On the other hand, this performs better than session-based approaches since no sessions are started every time a user logs in, in contrast to token-based authentication (Balaj 2017). The authors in a study (Rahmatulloh et al. 2018) have discussed REST security utilizing JWT authentication in his work.

3.1 Token authentication

Traditional authentication relies on session IDs stored on servers, which can become cumbersome and limit

Table 1 Comparative analysis of cloud security solutions

Source	Year	Proposal	Parameters	Data encryption technique	Key encryption technique	A	C	I	DAC
Mudepalli et al. (2017)	2016	A practical method for retrieving ciphertext from enormous amounts of data	Communication cost	RSA	ECC	×	√	×	×
Ahmad et al. (2024)	2017	A technique using a hybrid crypto system to store information in the cloud safely	–	RSA	RSA	×	√	√	×
Pravin et al. (2019)	2017	An effective system for sharing data securely in a multi-cloud environment that offers safe transactions and the adaptability of inter-cloud communication	Latency	3DES	ECC	×	√	√	×
Viswanath and Krishna (2021)	2020	A novel encryption technique in multi-cloud that combines the functionality of AES and Feistel network functionality for securing big data	Throughput running time	AES	Fiestal	×	√	×	×
Ismail et al. (2020)	2020	An architecture for implementing flexible, scalable, and fine-grained data storage and access management outsourcing	–	RSA	RSA	×	√	√	√
Shivaramakrishna and Nagaratna (2023)	2023	This work proposes a new Heterogeneous Cryptographic Model for cloud data storage security that combines TLAC, AKM and RSA with AES-OTP encryption technique	Accuracy, Precision, Recall, and F1-Score	AES	RSA	√	√	√	×
Konwar et al. (2024)	2024	The goal of this work is employing two factor authentication for cloud applications to enhance the security of the user credentials and the OTP to protect the privacy of data	QR code	SHA—256	RSA	×	√	√	×
Ahmad and Mehfuz (2024)	2024	Traditional encryption techniques have problems about the inability to resist tampering and the efficiency of encryption. This article provides TLADE, a more effective encryption algorithm than the existing ones	Quality of Service support values, latency factor	TLADE data encryption	TLADE data decryption	√	√	√	×
Proposed approach	2024	A TOTP-based secure data storage system in the cloud environment using the JWT Token approach	Time is taken for encryption and decryption	AES	RSA	√	√	√	√

scalability. Token-based authentication solves this problem. Tokens are secure, self-contained units containing user information that eliminate the need for session IDs on the server. This approach reduces server load, simplifies user interaction, and works well for distributed and cloud-based systems.

A person's username and password can be used to obtain tokens through token-based authentication, giving them access to specific resources. After resources become accessible, users can access them with the same tokens repeatedly (Bhawiyuga et al. 2017) without entering their username and password each time. For us to identify who owned the tokens and what access rights belonged to the user, the server created the given token using a certain technique. JWT is just a piece of JSON-formatted data that has been signed. A public and private key pair for RSA, ECDSA, or an HMAC symmetric key technique can be used to sign JWT (Jones et al. 2015). Recent studies simply compare the efficiency of the HMA SHA-256 algorithm against that of the SHA-512 method. The possible algorithm alternatives for JWT have yet to undergo performance testing (Rahmatulloh et al. 2018).

Figure 1 depicts the traditional client-server authentication. In contrast, a token that can be applied to future requests is swapped for the user's credential token in modern authentication. Users can be associated with tokens via cookies, local storage, or browser cookies.

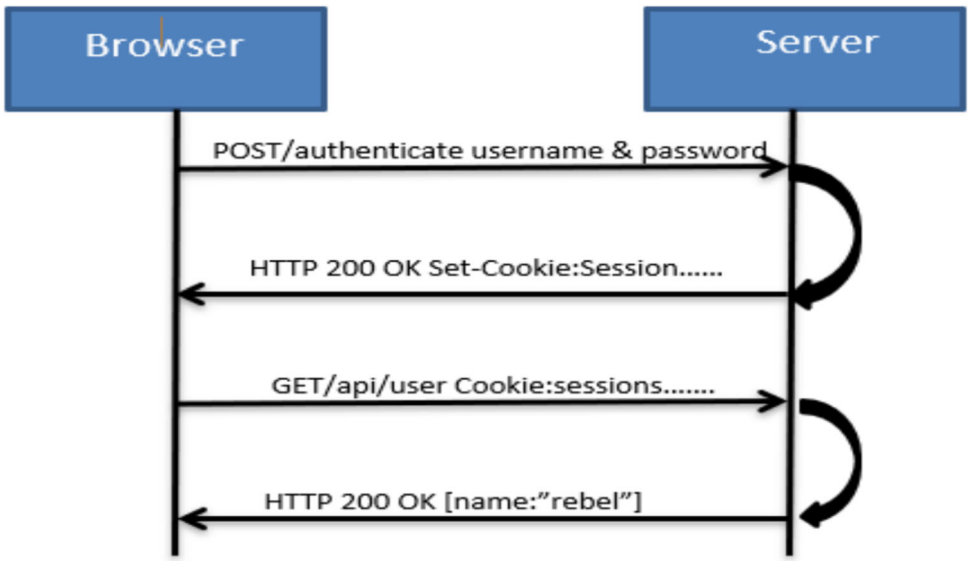


Fig. 1 Traditional cookie-based authentication



Fig. 2 Three main elements of JWT

3.2 JSON web token (JWT)

JWT, sometimes known as the "jot," is a token in a string form. It comprises three components: the header, payload, and signatures. It is employed in processes involving the exchange of information and authentication (Jones et al. 2015). Figure 2 shows a dot symbol (.) separating each component.

There are two main categories of tokens used for authentication: Informational Tokens, like identity tokens, tell the system who you are but don't grant access to specific resources (Zheng and Jiang 2014). Think of them as ID cards. Access Tokens, on the other hand, function like keys that unlock specific resources or functionalities within the system. JWTs (JSON Web Tokens) are a common example of access tokens. Once you log in successfully, the server provides a JWT stored in your browser cookies. This token acts as your "key" to access various parts of the system until it expires. In Fig. 3, we can see the JWT structure. The algorithm used and the token type are both included in the header section. After that, JSON is Base64 encoded. The claim, which is a statement about the entity is typically made up of additional users and data, is contained in the payload, which is the second component. The payload is identical to the Base64-encoded header. The third component is the signature which includes the sign, the secret, the algorithm defined in the header, the encoded payload, and the header.

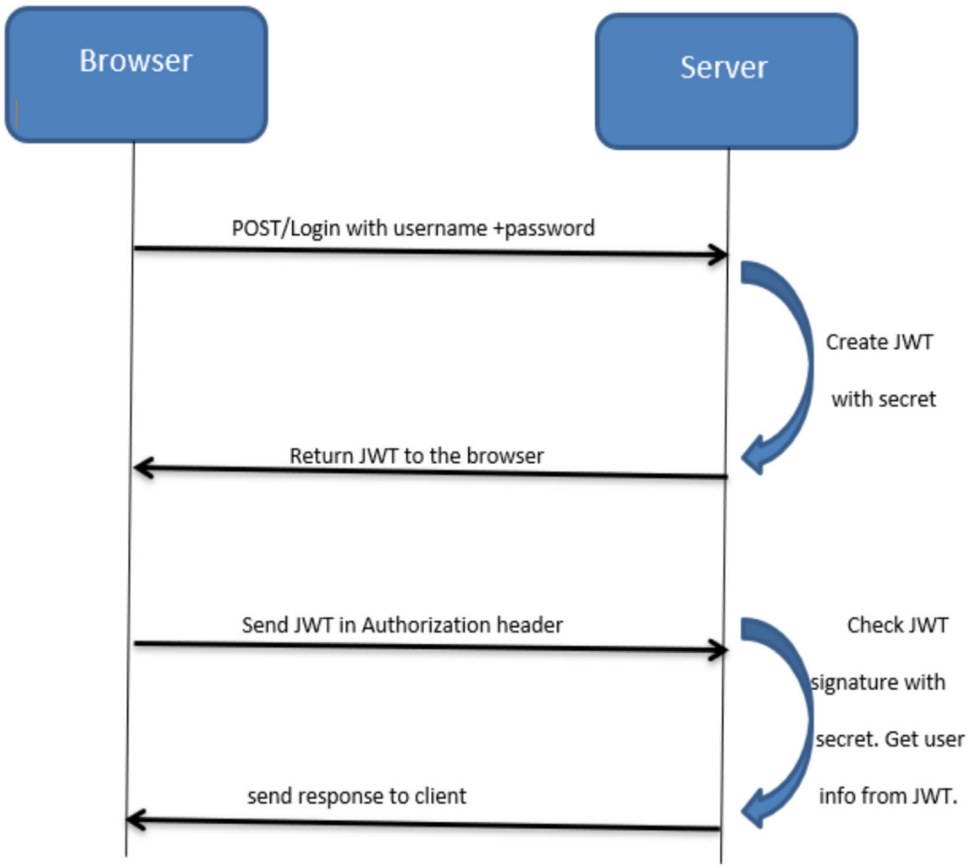


Fig. 3 JWT-based token authentication

```
Header: Algorithms and Token Types
    {
    "alg": "HS256", // Algorithm used for hashing
    "typ": "JWT" // Type of token (JSON Web Token)
    }
Payload: Data
    {
    "iss": "#appbackend", // Issuer information
    "user": "shahnawaz", // User data
    "pass": "Shah@123" // Password data
    }
Signature: results from Hash
    {
    "Base64-encoded (Header.Payload)" + "key" + "Algorithm" // Signature computation
    }
```

Because the JWT serves as a password, the server issues a token when a user successfully signs in. Then, additional requests are made to the server using the token. The primary goal of JWT is to safeguard the exchange of data between systems by requiring tokens in client requests for data sharing. Figure 3 illustrates the typical operational flow of JWT mechanisms.

3.3 User registration/authentication

To store data securely on this application, the user should register their information during the registration phase. Then, the registered information can be verified during the authentication phase. Using a login/password and an OTP, this phase ensures all users are authenticated and registered, as shown in Fig. 4. The following steps outline the data flow for registration:

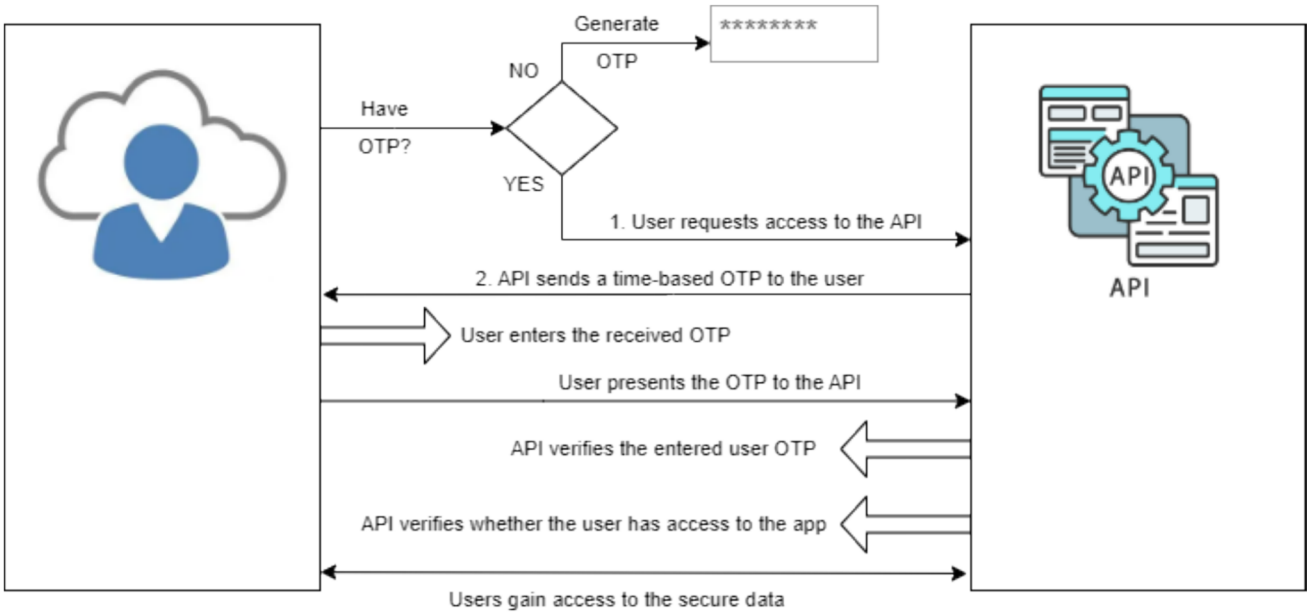


Fig. 4 Authentication/registration process

User→Get registered: Username/Email Address, Password, Forget Password
Registered User: Pub_{Key}, $Priv_{Key}$
$Password_{Enc}$ ($Priv_{Key}$)
otp←generatedotp()
}

After registration, a cloud user can log in and use the Cloud Documents program which is explained in detail in Sect. 4.

3.4 TrustStore

The TrustStore architecture is depicted in Fig. 5. It is a secure cloud storage platform that guarantees cloud-stored data privacy, availability, and confidentiality (Yao et al. 2010). Its main characteristics are:

- The current cloud storage services offer various storage services from multiple cloud vendors as an overlay storage solution. Furthermore, hybrid, redundant virtual file systems are created to achieve high data availability.
- The keys can be decoupled from the encrypted data using a separate KMS which can be easily installed in a (semi)-trusted environment.
- It provides a service for unbiased integrity management that facilitates verification of both online and offline integrity (Nepal et al. 2011).
- It provides a wide range of APIs to enable the development and implementation of multiple applications across numerous platforms.

TrustStore operates in three main layers: the client application, the server-side services that manage the system, and the cloud storage where data is kept. It was constructed as per service-oriented architectural concepts. As a result, each layer's services are created and implemented independently, interacting via a defined API. As a result, different applications using TrustStore can be developed and deployed more flexibly because of the services' loose coupling. TrustStore saves all data in encrypted form on all cloud services by default and treats it all as sensitive data. Additionally, made possible by this multi-cloud capability is the high availability needed for a safe system.

3.5 Servers for TrustStore

According to the guidelines of service-oriented architecture, the TrustStore Servers are implemented as services. These services run in a reliable setting. Two essential services—confidentiality and integrity—are created to support two crucial aspects of data security. At the core of the server-side software lies the KMS. It handles several crucial tasks: (1) securing data encryption keys, (2) managing user login credentials, and (3) controlling how users share data within the system. These features address data security and sharing, like the goals of cloud

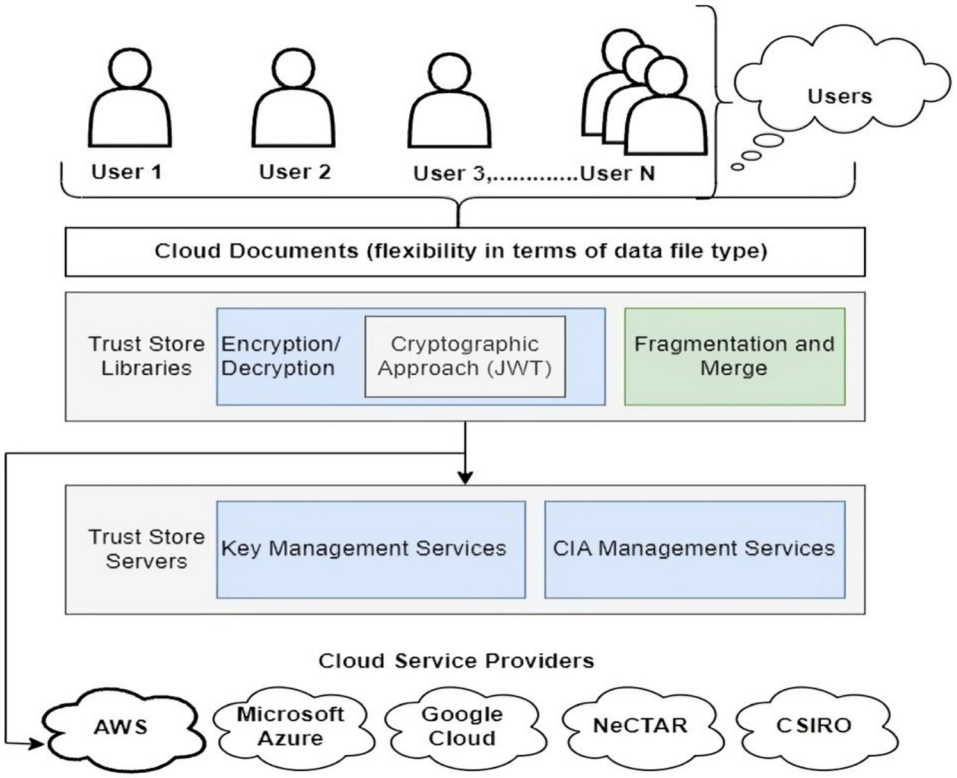


Fig. 5 Schematic diagram of the TrustStore

documents (Uymatiao and Yu 2014a). Additionally, the Integrity Management Service (IMS) verifies that data remains unaltered, ensuring its integrity.

The system achieves robust security through three key services: KMS safeguards data privacy with encryption, IMS ensures data remains unaltered, and client replication provides redundancy for additional protection.“ Server-side services can be offered by reputable third parties and are not necessarily provided by TrustStore. For the protection of data with higher or lower security, additional KMS instances might be built in mostly secure environments, or data could be encrypted and stored in numerous storage services. If an organization’s employees use it, the IT department can offer dependable services to all employees rather than requiring that each employee have their services.

3.6 TrustStore client

The TrustStore Client acts as your bridge to the system. It provides a user-friendly interface for interacting with TrustStore. Additionally, it handles security behind the scenes, including partitioning data locally, replicating it for backup, and encrypting/decrypting your files using secure methods. To achieve this, the client relies on powerful cryptographic libraries like AES, RSA, CMS, and SHA.

The core capabilities of the TrustStore client can be boiled down to the following.

When a file is uploaded by the user, it is split up into several sections initially. After that, a fragmentation map is made in which the encryption/decryption module individually encrypts the fragmented data. Finally, a fragmentation map is used to piece the various fragments back together during download to create the entire file.

To ensure top-notch security, TrustStore breaks down your data into encrypted fragments. Each fragment has its own unique key, generated randomly, to keep things extra secure. These keys are safely stored in the KMS. The encrypted fragments are then uploaded to cloud storage. When you download your data, the system retrieves the pieces from the first available storage provider and decrypts them using the corresponding keys retrieved from the KMS. This system builds upon the secure data sharing features of Cloud Documents (Wise et al. 2015) by managing user accounts, access controls, and encryption keys for all your data and its associated information. TrustStore offers secure cloud storage services for a variety of application areas. As a trustworthy integration of TrustStore, we have created an application dubbed TOTP-based secure data storage solution. The post-genomic life sciences have served as the inspiration for this study (Sinnott et al. 2013; Ahmad et al. 2023). In this concept, the cloud stores and analyzes sensitive data, such as human genomic data, to find disease-causing variations. The thorough analysis of huge data sets necessitates defining and developing intricate workflows that handle the

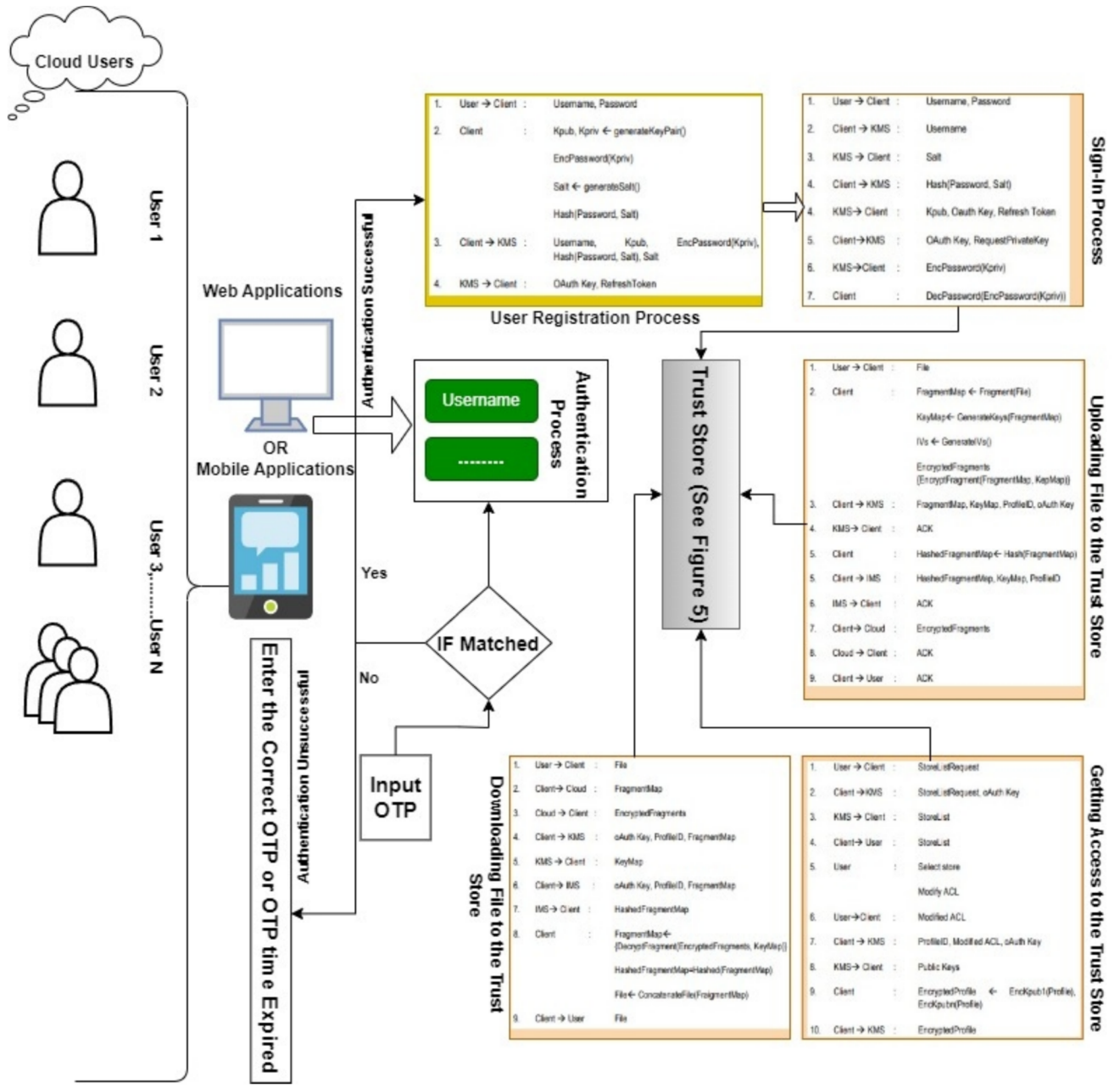


Fig. 6 Proposed system architecture

numerous privacy and security considerations concerning such data. The TruXy system (Nepal et al. 2011), which focuses on the workflow environment, was created to address these difficulties. In addition, the necessity for safe data sharing and collaboration is a typical demand. As a result, we created Cloud Documents, a generic application that functions as a TOTP-based secure data storage system for data sharing. In the following sections, we'll review the Cloud Documents system's protocols and how it operates.

4 Proposed secure data storage system in the cloud

This section unveils a novel architecture for safeguarding data across multiple cloud storage platforms. Our unique system merges two established encryption methods: TOTP and the powerful Advanced Encryption Standard (AES). TOTP adds an extra layer of security like online banking codes, requiring a frequently changing temporary code for access. AES scrambles your data using different key sizes (128-bit, 192-bit, and 256-bit) and a larger block size (128 bits) for enhanced protection. By combining these

methods, our approach offers robust security for your data within a multi-cloud environment.

The suggested framework's primary goals are:

4.1 Ensure quick, light, and secure data storage

AES encryption outperforms other algorithms in processing speed, as illustrated in Mota et al. (2018).

According to the experimental findings of another study (Mushtaq et al 2017), AES can also be used in applications where integrity and confidentiality are of utmost importance because it uses less power and executes programs faster than other symmetric algorithms. Therefore, due to the AES method's fast encryption/decryption times, we incorporated it into our solution.

4.2 Ensure secure and safe key exchange

Despite being slower, the asymmetric method produces good results for safe data transfer, ensuring secure key exchange. Hence, we have encrypted the AES algorithm's secret key using RSA. Once you encrypt something with a public key, you need the private key that goes with it to get it back.

4.3 Ensure reliable and robust authentication

For safe, protected, and reliable access from the cloud, a TOTP is generated over email. The popular SMS OTP authentication method has yet to be used in our implementation because it has many drawbacks. Recent research has shown that OTPs sent by SMS are responsible for various problems that cause waste of time and delays in various processes (Viswanath and Krishna 2021; Dhanashri Ghosalkar 2019). Furthermore, SMS OTP may experience financial difficulties if the carrier charges the subscriber for SMS messaging. The registration generated TOTP encrypts and protects the data inside the system against misuse.

Table 2 Analysis of the upload times for RSA and AES

File (Extension)	Size of file (KB/MB)	Uploading time (in ms)	
		RSA	AES
.txt	7 KB	0.06	0.06
.mp4	700 KB	0.1	0.07
.pdf	2.1 KB	0.22	0.12
.mp3	5 MB	0.88	0.56
.mp3	10 MB	0.81	0.75
.mp4	86.4 MB	7.45	6.78
.pdf	177 MB	15.69	10.9

Figure 6 depicts a high-level diagram of the TOTP enrollment procedure.

Figure 6 displays a generalized representation of the proposed architecture. The first part of our solution, the registration and authentication process, consists of three main steps. First, it guarantees that every user is identified correctly and registered inside our system. The second part of the procedure, trust storage, allows users to upload and download files of any extension with no file size limitations. Finally, the process of recovering a server is the third phase. By entering a new username and matching the password, a new user can register to utilize a cloud document. The user is registered with KMS, after which the client creates a public–private key pair. The public key is stored in an X.509v3 RSA certificate, and the private key is encrypted using PKCS#8 EncryptedPrivateKeyInfo along with a password. The PBKDF2 algorithm with SHA256 is used in this process. All communications with KMS, as depicted in steps 3 and 4 in Fig. 6, are conducted exclusively over SSL. The password is necessary to decrypt the user's private key, which is used to access any data stored in cloud documents. This password is never shared with KMS or any other services. Therefore, even if KMS, IMS, and cloud storage systems are compromised or collaborate, the data remains secure and inaccessible.

Signing into the cloud documents program prioritizes security. First, you provide your username and password. The system then retrieves a secret code (login salt) from a secure KMS for an extra layer of protection. It combines your password with this code and sends it to the KMS for verification. If everything matches, the KMS sends back an encrypted version of your private key, which is crucial for decrypting data. Finally, you use your original password to unlock this private key and gain access. It's important to note that everyone has access to public keys, which are used for encrypting data but not decrypting it.

Uploading a file in the cloud documents program prioritizes both security and redundancy. After selecting a file, the system breaks it down into smaller pieces for secure storage. Each piece gets encrypted with a unique key and initialization vector for added protection. These encrypted pieces, along with their keys and vectors, are then sent to a secure key management system for safekeeping. To ensure data integrity, a unique fingerprint (hash) is created for each encrypted piece and stored separately. Finally, the encrypted pieces are uploaded to all available cloud storage providers, guaranteeing access even if one provider becomes unavailable. This multi-step process safeguards your data and offers peace of mind.

Sharing stores within the cloud documents program revolves around secure access control. First, locate the store you want to share from the list. Remember, only stores where you hold administrative, or ownership rights

can be modified. To grant access to others, the system retrieves their public keys from a secure KMS. These public keys are then used to encrypt the user profile with a method called SMIME encryption. This encrypted profile is then stored in KMS for authorized users to access. When someone attempts to open a shared store, the client retrieves the store credentials from KMS, which were previously encrypted using the user's public key. Finally, the client leverages the user's private key to decrypt the credentials and access the store's data on the cloud storage providers. This ensures only authorized users can access shared stores, with their access tightly controlled through secure encryption methods.

The user can choose a file to open from the list of files in the storage. The client must download the pieces and put them back together to support this. The codenames for the relevant fragments are located by searching the file system. The client then asks KMS for the decryption keys. Finally, the system retrieves the file fragments for download. It checks any available cloud storage provider using the credentials stored in the profile. If a particular provider is unavailable for some reason, the system automatically attempts to download from the next available storage provider, ensuring redundancy and uninterrupted access to your data.

5 Performance analysis and assessment

In this section, we implemented our approach by storing and retrieving encrypted data using multiple clouds. The web app used for encrypting and decrypting data is coded in Java and hosted on AWS. First, we assess the upload time of our solution, as given in Table 2. To determine the method for the upload time, we varied the file types and sizes. We measure the time complexity of our solution, which takes uploading time into account.

The authors conducted a performance analysis by comparing their proposed authentication system with well-known methods and recent similar approaches. They assessed various metrics such as login success rates, data confidentiality, data integrity, and access control. Through this comparison, it was evident that their system achieved higher login success rates, provided stronger data confidentiality, ensured data integrity, and offered more robust access control mechanisms when compared to the alternatives. This empirical evidence supported the conclusion of better system performance in these critical aspects.

5.1 Experimental environment

Our findings from Table 2 demonstrate that using the AES algorithm results in shorter upload times compared to the RSA Algorithm. This suggests that minimal time was expended during the authentication phase.

- *Upload Time*: The amount of time required to encrypt, upload, and distribute the file to various Cloud Service Providers.
- *File type*: gif, svg, tif, docx, html, xls, mp4, mov, ppt, pdf, mp3, txt, mp4, png, jpeg (Files of any type can be uploaded—Our framework supports all file types)
- *File Size*: 7 KB–5 GB
- (The file sizes mentioned are for experimental purposes. Our framework supports files of substantial size.)

This includes measuring upload time for encrypting and distributing files across various Cloud Service Providers, supporting a wide range of file types, and accommodating file sizes from 7 to 5 GB, with the ability to handle larger files.

We observe from Table 2 that the upload time using the AES algorithm is lesser when compared to using the RSA Algorithm. This indicates that very little time was lost during the authentication phase.

5.2 Discussion and comparison

In this section, the authors implemented their approach for storing and retrieving encrypted data using multiple cloud platforms, with a Java-based web application hosted on AWS. They evaluated the system's performance, specifically focusing on upload times, which are crucial for data transfer and user experience.

The experimental environment was designed to measure upload times across various cloud service providers. It accommodated a wide range of file types and sizes, from 7 to 5 GB, and could handle larger files. The results, as shown in Table 2, compared the upload times between the AES and RSA algorithms.

The findings indicated that the AES algorithm outperformed the RSA Algorithm in terms of upload time. This observation suggests that the proposed system achieved a faster data upload process, which is significant for user authentication and system efficiency. The comparison of AES and RSA algorithms revealed that the AES algorithm, known for its strong encryption capabilities, was more efficient in this context. The shorter upload times with AES imply that the authentication phase, which involves data transfer, was more streamlined and faster, contributing to a smoother and more responsive user experience. In summary, the experiment highlighted the practical advantages of utilizing the AES algorithm for faster data upload and authentication, showcasing the system's superior performance in this critical aspect when compared to the RSA Algorithm.

6 Use case of the proposed framework

The proposed TOTP-based secure data storage system using JWT Token methodology is useful for the applications which requires high level of security such as banking applications. Such systems have operational and non-functional characteristics, which qualify them for this research implementation. The features of a bank for operation are customer login, funds' viewing, transfers, bill payments, cheques, and utility services. Some of the non-functional requirements include performance requirements that include increasing the number of products sold, decreasing costs, and safety features like backup and recovery procedures, security features like account ID and PIN and failed login attempts and encryption that requires algorithm such RSA, AES, or ECC. These are usability, availability, confidentiality, and integrity which are other non-functional requirement of system. Also, these banks are involved in different authentication chores including Internet transactions, preparation of challans, downloading of statements, whether monthly, yearly, or weekly, personal transaction history, secure money transfer, withdrawal of money and depositing of cash. Thus, the utilization of TOTP-based framework can improve security of these banking functions, protect important information, preserve privacy of users, and meet higher level of security requirements, so it can be considered as useful addition to banking systems.

7 Conclusion

Cloud storage is becoming more and more popular in a variety of fields, including healthcare, banking, governance, and defense, where confidential and sensitive data are being stored into the platform and thus requires higher degree of protection mechanism to ensure the confidentiality and integrity of data. A secure cloud storage framework called TrustStore has been proposed in the work. TrustStore provides two levels of protection mechanism one stops the unauthorized access of platform from illegitimate users and other ensures safety and security of data while in storage. A time-based OTP called TOTP has been implemented to ensure that only legitimate users will be able to have access to the systems. Additionally, RSA and AES have been implemented to ensure confidentiality and integrity of data in the vault. The proposed framework does not charge the user and is completely free. There will be zero SMS costs to the user and server because users will not require the SMS functionality. This technique could be used to enable multi-factor authentication in a variety of applications. The proposed TOTP-based framework offers a robust solution for addressing both operational and non-functional security requirements in banking systems, safeguarding critical data, and preserving user privacy during various authentication activities. This framework enhances security measures for online transactions, statement downloads, and other banking functions while ensuring confidentiality, integrity, and availability.

Acknowledgements The authors acknowledge the financial support received on their own, for their support and encouragement in carrying out his college work. The authors also would like to acknowledge the administration of Bennett University, Galgotias University, Sharda University, NIET, Jamia Millia Islamia, and University of 20 août 1955—Skikda which the authors represent.

Author contributions Conceptualization, SA, and SM; data curation, SH and MA and JA; formal analysis, MA, and JA; investigation, SM, MA, and SA; methodology, SA, and SM.; resources, MA, and JA; supervision SM; validation, SA, MA and JA; visualization, SM; writing—original draft, SA and MA; writing—review and editing, SA, MA and JA.

Funding Not applicable.

Data availability Not applicable.

Materials availability Not applicable.

Code availability Not applicable.

Declarations

Conflict of interests The author's declare that they have no conflict of interest.

Consent for publication Not applicable.

Ethics approval and consent to participate Not applicable.

References

Ahmad S, Mehfuz S (2024) Efficient time-oriented latency-based secure data encryption for cloud storage. Cyber Secur Appl 2:100027

Ahmad S, Mehfuz S, Beg J (2023) Hybrid cryptographic approach to enhance the mode of key management system in cloud environment. J Supercomput 79(7):7377–7413

Ahmad S, Mehfuz S, Urooj S, Alsubaie N (2024) Machine learning-based intelligent security framework for secure cloud key management. Cluster Comput 27(5):5953–5979

Aigbe P, Akpojaro J (2014) Analysis of security issues in electronic payment systems. Int J Comput Appl 108(10):10–14

Aina F, Yousef S, & Osanaiye O (2018) Design and implementation of challenge response protocol for enhanced e-commerce security

Alam SS, Ali MH, Omar NA, Hussain WMHW (2020) Customer satisfaction in online shopping in growing markets: an empirical study. Int J Asian Bus Inf Manag 11(1):78–91

Alsayed AO, Bilgrami AL (2017) E-banking security: internet hacking, analysis and prevention of fraudulent activities. Int J Emerg Technol Adv Eng 7(1):109–115

Arvind A, Mahajan P, Chalke R (2020) TOTP based authentication using QR code for gateway entry system. Int J Eng Comput Sci 9(05):25023–25028

Balaj Y (2017) Token-Based vs. Session-Based Authentication: A survey 1–5

Bansal S, Aggarwal H (2023) A hybrid particle whale optimization algorithm with application to workflow scheduling in cloud–fog environment. Decis Anal J 9:100361

Bansal S, Aggarwal H (2024a) An efficient workflow scheduling in cloud-fog computing environment using a hybrid particle whale optimization algorithm. Wirel Pers Commun 137:441–475. https://doi.org/10.1007/s11277-024-11421-8

Bansal S, Aggarwal H (2024b) A multiobjective optimization of task workflow scheduling using hybridization of PSO and WOA algorithms in cloud-fog computing. Cluster Comput 27:10921–10952. https://doi.org/10.1007/s10586-024-04522-3

Bhawiyuga A, Data M and Warda A (2017) Architectural Design of Token based Authentication of MQTT Protocol in Constrained IoT Device, In: 2017 11th International Conference on Telecommunication Systems Services and Applications (TSSA), 1–4

Cai X, Yuen C, Huang C, Xu W, Wang L (2023) Toward chaotic secure communications: an RIS enabled M-ary differential chaos shift keying system with block interleaving. IEEE Trans Commun 71(6):3541–3558. https://doi.org/10.1109/TCOMM.2023.3262834

Chen K (2023) Confidential high-performance computing in the public cloud. IEEE Internet Comput 27(1):24–32. https://doi.org/10.1109/mic.2022.3226757

Choudhary A, Rajak S, Shinde A, Warkhade S, Ghodichor FS (2017) Online banking system using mobile-OTP with QR-code. Int J Adv Res Comput Commun Eng 6(4):657

Dhanashri Ghosalkar SP (2019) OTP over SMS: Time delay issues and causes," pp 1–7

Namasudra S, Nath S and Majumder A (2014) Profile based access control model in cloud computing environment, In: 2014 International Conference on Green Computing Communication and Electrical Engineering (ICGCCEE), Coimbatore, India, 2014, pp 1–5, https://doi.org/10.1109/ICGCCEE.2014.6921420.

Fatima S, Rehman T, Fatima M, Khan S, Ali MA (2022) Comparative analysis of aes and rsa algorithms for data security in cloud computing. Eng Proc 20(1):14. https://doi.org/10.3390/engproc2022020014

Gualdoni J, Kurtz A, Myzyri I, Wheeler M, Rizvi S (2017) Secure online transaction algorithm: securing online transaction using twofactor authentication. Proc Comput Sci 114:93–99

Hassan MA, Shukur Z, Hasan MK (2020) An efficient secure electronic payment system for E-commerce. Computers 9(3):66

Hassan M A and Shukur Z (2019) Review of digital wallet requirements. In: 2019 International Conference on Cybersecurity (ICoCSec) pp 43-48 IEEE

Idayathulla R (2019) Enhanced adaptive security system for SMS – based One Time Password 5(4): 538–541

Islam S, Hashim AHA, Habaebi MH, Hasan MK (2017) Design and implementation of a multihoming-based scheme to support mobility management in NEMO. Wirel Pers Commun 95(2):457–473

Ismail T, Touati H, Hajlaoui N, Hamdi H (2020) Hybrid and Secure E-Health Data Sharing Architecture in Multi-Clouds Environment. In: Jmaiel, M., Mokhtari, M., Abdulrazak, B., Aloulou, H., Kallel, S. (eds) The Impact of Digital Technologies on Public Health in Developed and Developing Countries. ICOST 2020. Lecture Notes in Computer Science, vol 12157. Springer, Cham. https://doi.org/10.1007/978-3-030-51517-1_21

Jones M, Sakimura N and Bradley J 2015 RFC 7519: JSON Web Token (JWT) (https://www.heise.de/netze/rfc/rfcs/rfc7519.shtml)

Jumani AK, Laghari RA (2021) Review and state of art of fog computing. Arch Comput Methods Eng 28(5):3631–3643

Konwar R, Jha D, Agrawal R, Purkayastha R, & Banerjee I (2024) A two-factor authentication mechanism using a novel OTP generation algorithm for cloud applications. In: 2024 14th International Conference on Cloud Computing, Data Science & Engineering (Confluence) (pp. 245–250). IEEE.

Laghari AA, He H, Khan A, Laghari RA, Yin S, Wang J (2022) Crowdsourcing platform for QoE evaluation for cloud multimedia services. Comput Sci Inform Syst 19:38–38

Laghari AA, Zhang X, Shaikh ZA, Khan A, Estrela VV, Izadi S (2023) A review on quality of experience (QoE) in cloud computing. J Reliab Intell Environ 10:107

May L, Henricksen M, Millan W, Carter G, & Dawson E (2002) Strengthening the key schedule of the AES. In: Information Security and Privacy: 7th Australasian Conference, ACISP 2002 Melbourne, Australia, July 3–5, 2002 Proceedings 7 (pp. 226-240). Springer Berlin Heidelberg

Mohamad MSA, Din R, Ahmad JI (2021) Research trends review on RSA scheme of asymmetric cryptography techniques. Bull Electr Eng Inform 10(1):487–492. https://doi.org/10.11591/eei.v10i1.2493

Mota A V, Azam S, Shanmugam B, Yeo K C, & Kannoorpatti K (2018) Comparative analysis of different techniques of encryption for secured data transmission. In: IEEE International Conference on Power, Control, Signals, and Instrumentation Engineering, ICPCSI 2017 (Vol. 1, pp. 231–237). IEEE, Institute of Electrical and Electronics Engineers. https://doi.org/10.1109/ICPCSI.2017.8392158

Mudepalli S, Rao V S, & Kumar R K (2017) An efficient data retrieval approach using blowfish encryption on cloud ciphertext retrieval in cloud computing. In: 2017 international conference on intelligent computing and control systems (ICICCS) pp 267-271 https://doi.org/10.1109/ICCONS.2017.8250724.

Mushtaq M, Jamel S, Disina A, Pindar Z, Shakir N, Deris M, Mustafa. (2017) A survey on cryptographic encryption algorithms. Int J Adv Comput Sci Appl 8:333–344. https://doi.org/10.14569/IJACSA.2017.081141

Namasudra S, Devi D, Choudhary S, Patan R, & Kallam S (2018) Security, privacy, trust, and anonymity. In: Advances of DNA computing in cryptography (pp 138-150). Chapman and Hall/CRC. ISBN: 9781351011419, 2018

Nechvatal J, Barker E, Bassham L, Burr W, Dworkin M, Foti J, & Roback E (2001) Report on the development of the advanced encryption standard (AES). J Res Nat Inst Stand Technol 106(3):511

Nepal S, Chen S, Yao J, & Thilakanathan D (2011) DIaaS: Data integrity as a service in the cloud. In: 2011 IEEE 4th International Conference on Cloud Computing pp 308–315). IEEE

Pavlovski C, Warwar C, Paskin B and Chan G (2015) Unified Framework for Multi-factor Authentication, 22nd International Conference on Telecommunications (ICT 2015) 209–13

Pravin A, Jacob P, Nagarajan G (2019) A robust technique for data security in multi-cloud storage using dynamic slicing with hybrid cryptographic technique. J Ambient Intell Humaniz Comput. https://doi.org/10.1007/s12652-019-01563-0

Rahmatulloh A, Sulastri H, Nugroho R (2018) Keamanan RESTful Web Service Menggunakan JSON Web Token (JWT) HMAC SHA-512. J Nasional Teknik Elektro Dan Teknologi Informasi 7(2):131–137

Rani E, Bansal S, Tyagi A (2019) Usage of biometric security in web based architecture. Think India J 22(16):2546–2555

Rasori M, Manna ML, Perazzo P, Dini G (2022) A survey on attribute -based encryption schemes suitable for the internet of things. IEEE Internet Things J 9(11):8269–8290. https://doi.org/10.1109/jiot.2022.3154039

Shivaramakrishna D, Nagaratna M (2023) A novel hybrid cryptographic framework for secure data storage in cloud computing: integrating AES-OTP and RSA with adaptive key management and time-limited access control. Alex Eng J 84:275–284

Sinnott R O, Bruns L, Duran C, Hu W, Jayaputera G, & Stell A (2013) Development of an endocrine genomics virtual research environment for Australia: building on success. In: Computational Science and Its Applications–ICCSA 2013, pp 364–379

Stallings W (2005) Cryptography, and network security: principles and practices. Prentice Hall

Taher K A, Nahar T, & Hossain S A (2019). Enhanced cryptocurrency security by time-based token multi-factor authentication algorithm. In: 2019 International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST) pp 308-312 IEEE

Uymatiao M L T and Yu W E S (2014). Time-based OTP authentication via secure tunnel (TOAST): A mobile TOTP scheme using TLS seed exchange and encrypted offline keystore. In: 2014 4th IEEE International Conference on Information Science and Technology pp 225-229 IEEE.

Uymatiao M L T, & Yu W E S (2014). Time-based OTP authentication via secure tunnel (TOAST): A mobile TOTP scheme using TLS seed exchange and encrypted offline keystore. In: 2014 4th IEEE International Conference on Information Science and Technology (pp. 225-229). IEEE

Viswanath G, Krishna P (2021) Hybrid encryption framework for securing big data storage in a multi-cloud environment. Evolut Intell. https://doi.org/10.1007/s12065-020-00404-w

Wise C, Friedrich C, Nepal S, Chen S, & Sinnott R O (2015). Cloud Docs: secure scalable document sharing on public clouds. In: IEEE 8th International Conference on Cloud Computing (CLOUD), pp 532–539.

Xuanzhi L and Ahmad K (2019). Factors affecting customers satisfaction on system quality for e-commerce. In: 2019 International Conference on Electrical Engineering and Informatics (ICEEI) pp 360-364 IEEE

Yao J, Chen S, Nepal S, Levy D, & Zic J (2010) Truststore: Making amazon s3 trustworthy with services composition. In: Proceedings of the 10th IEEE/ACM International Conference on Cluster, Cloud and Grid Computing, pp 600–605

Zheng K, Jiang W (2014) A token authentication solution for Hadoop based on Kerberos Preauthentication. Int Conf Data Sci Adv Anal 2014:354–360

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.